TOPIC 02 · DATA ANALYTICS AND VISUALIZATION
LAB 7 THEME · DATA ANALYSIS & INSIGHTS

Lab 7 — Generate Hiring Insights and Pivot Tables with Gemini

Maps to	LO2: use Gemini to analyse an HR dataset - trends, comparisons and rates - inspect its analysis steps, create and adjust pivot tables, verify every number with formulas and export the insights to Google Docs
Duration	30 minutes
Tools	Google Sheets · Gemini data analysis (Analysis steps, Export to Docs) · pivot-table actions · OrchidBay_Hiring_2026_YTD.xlsx
Mock data	data/OrchidBay_Hiring_2026_YTD.xlsx
Course	Gemini for Google Sheet (C817) · v2.0
Prompts	https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-7

Goal

Answer the Head of Talent's questions about 2026 hiring - trend, best source, slowest role, cost per hire - with numbers you have checked.

What You Will Do

Orchid Bay's Head of Talent wants to know how 2026 hiring is going. Using all 720 applications from January to September, you will ask Gemini what analysis is possible, find the monthly trend, compare sources by hire rate and cost per hire, and find which role takes longest to fill. You will open Analysis steps to see how Gemini calculated, create pivot tables with Gemini actions and adjust them with Settings, check every figure with COUNTIFS or AVERAGEIFS, and export five insights to a Google Doc.

What You Will Build

OrchidBay_Hiring_2026_YTD with two pivot tables (source by stage, days to hire by position), a Checks area of formulas, and a Google Doc of insights exported from Gemini

Prerequisites

- Labs 1-6, or comfort with Ask Gemini, Insert and actions.
- This lab folder downloaded, including data/OrchidBay_Hiring_2026_YTD.xlsx.

Data note

Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Steps

Step 1. Download OrchidBay_Hiring_2026_YTD.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

PROMPT / CHECK — copy and adapt

```
Required state:
File : OrchidBay_Hiring_2026_YTD (Google Sheets copy)
Tabs : About_This_Data | Applications (720 rows) | Jobs | Source_Costs | Prompt_Log
Period : applications 2026-01-01 to 2026-09-26
```

Step 2. Start broad. Gemini suggests analyses that fit the columns it finds - note two you want to try.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
What analysis can you help me perform with this sheet?
```

Step 3. Ask for the trend. Under the answer, click Analysis steps (and Show code if offered) to see how Gemini grouped and counted. Applications peaked in June (98); February was lowest (62).

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Identify trends in the number of applications by Month. Which month had the most applications,
and which had the fewest?
```

Step 4. Compare sources. Gemini should find that Referral has the best hire rate (10 of 78, 12.8%) while LinkedIn brings the most hires (11 of 184).

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
For each Source, show the applications, hires (Stage is Hired) and hire rate. Which source has
the best hire rate, and which brings in the most hires?
```

Step 5. Create a pivot table with a Gemini action. When it finishes, click Settings to open the pivot table editor - try adding a filter. LinkedIn should show Hired 11 and Rejected 147; the grand total is 720. Then paste the checks into empty cells beside the pivot table.

PROMPT / CHECK — copy and adapt

```
ACTION:
Create a pivot table on a new sheet that shows the count of Candidate_ID with Source in rows
and Stage in columns.

CHECK:
LinkedIn hired =COUNTIFS(Applications!G2:G721,"LinkedIn",Applications!M2:M721,"Hired")
Referral hired =COUNTIFS(Applications!G2:G721,"Referral",Applications!M2:M721,"Hired")
All applications =COUNTA(Applications!A2:A721)
```

Step 6. Which role is slowest to fill? Gemini should name HR Executive (39.3 days on average; 31.1 days across all 41 hires). Confirm with a second pivot table and an AVERAGEIFS check.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
What is the average Days_To_Hire for each Position? Which role takes the longest to fill?

ACTION:
Create a pivot table that shows the average Days_To_Hire by Position, sorted from highest to
lowest.

CHECK:
HR Executive avg =AVERAGEIFS(Applications!P2:P721,Applications!F2:F721,"HR
Executive",Applications!M2:M721,"Hired")
```

Step 7. Bring in costs from a second tab. Cost per hire = applications x cost per application / hires. Company Website should be cheapest (S$125) and Career Fair dearest (S$1,540).

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Using the Applications and Source_Costs tabs, calculate the total cost and the cost per hire
for each Source. Which source gives the cheapest hires?
```

Step 8. Ask for insights for the Head of Talent. Check each number against your pivot tables and checks, then click Export to Docs and name the document 'Orchid Bay 2026 Hiring Insights'. Optional: ask the web question - Gemini only searches the web when you ask it to, and you must check its sources.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Write five insights for the Head of Talent from this data. Give the number behind each insight
and one recommended action.

ASK GEMINI:
Using Google Search, what is a typical time to hire for professional roles in Singapore? List
your sources.
```

Step 9. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

PROMPT / CHECK — copy and adapt

```
LOG:
List the prompts I sent you in this conversation. For each one, give a one-line summary of
what you did, so I can copy it into the Prompt_Log tab.
```

Sample Prompts

Use these prompts in the Gemini side panel in Google Sheets (or adapt them). Name the tab and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online: https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-7

#	Purpose	Prompt
1	Trend	How did the number of hires change from Q1 to Q2 to Q3 2026?
2	Compare	Compare the average Screening_Score of hired and rejected candidates.
3	Pivot	Create a pivot table that shows the number of hires by Recruiter and Month.
4	Regression	How can I show regression and prediction of monthly applications?
5	Explain	Explain this analysis like I'm a hiring manager with no data background.

Test It

Check	Expected result
Trend	Peak June 2026 (98 applications); lowest February (62).
Sources	Best hire rate Referral 12.8%; most hires LinkedIn (11); 41 hires in total.
Pivot 1	Source x Stage; LinkedIn Hired 11, Rejected 147; grand total 720.
Pivot 2	HR Executive slowest at 39.3 days; AVERAGEIFS matches.
Cost per hire	Cheapest Company Website S$125; dearest Career Fair S$1,540; insights exported to a Google Doc.

Checkpoint

Lab 8 charts the same data from its own copy of the workbook.

Troubleshooting

If this happens	Fix
No Analysis steps button.	It appears only for answers Gemini calculated from the sheet. Ask for a calculation, e.g. 'count by Month'.
Gemini counts 721 applications.	It included the header row. Ask it to count data rows only, or check with =COUNTA(Applications!A2:A721).
Hire rate differs from yours.	Ask how it defined a hire. It must be Stage = Hired, divided by all applications from that source.
No action card for the pivot table.	Build it from the menu: select the data > Insert > Pivot table > New sheet, then add Source to Rows, Stage to Columns and COUNTA of Candidate_ID to Values.
Export to Docs is missing.	It is shown for analysis responses. Use Copy, then paste into a new Google Doc.
The web answer has no sources.	Do not use it. Ask again with 'list your sources', or use Retry with Google Search.

Challenge

Ask Gemini whether candidates with a higher Screening_Score are more likely to be hired, and check its claim with two AVERAGEIFS.

Reflection

Referral has the best hire rate but only 78 applications. What would you check before telling the business to 'do more' of it?